

updateProductQuantity

SYSTEM CONTRACT

Operation

```
updateProductQuantity(productId: String,  
newQuantity: int)
```

Cross References

Use Case c (Update inventory & alert shortages)

Pre-conditions

- A Product matching `productId` exists in the inventory.
- `newQuantity` is a non-negative integer.

Post-conditions

- The **ProductDL** instance's quantity attribute was updated to `newQuantity` .
- If `newQuantity` was strictly less than the product's minimum threshold, a new **LowStockAlert** instance *was created (instance creation)*.
- The updated ProductDL and the LowStockAlert (if created) *were saved* persistently to the database via InventoryDAO.